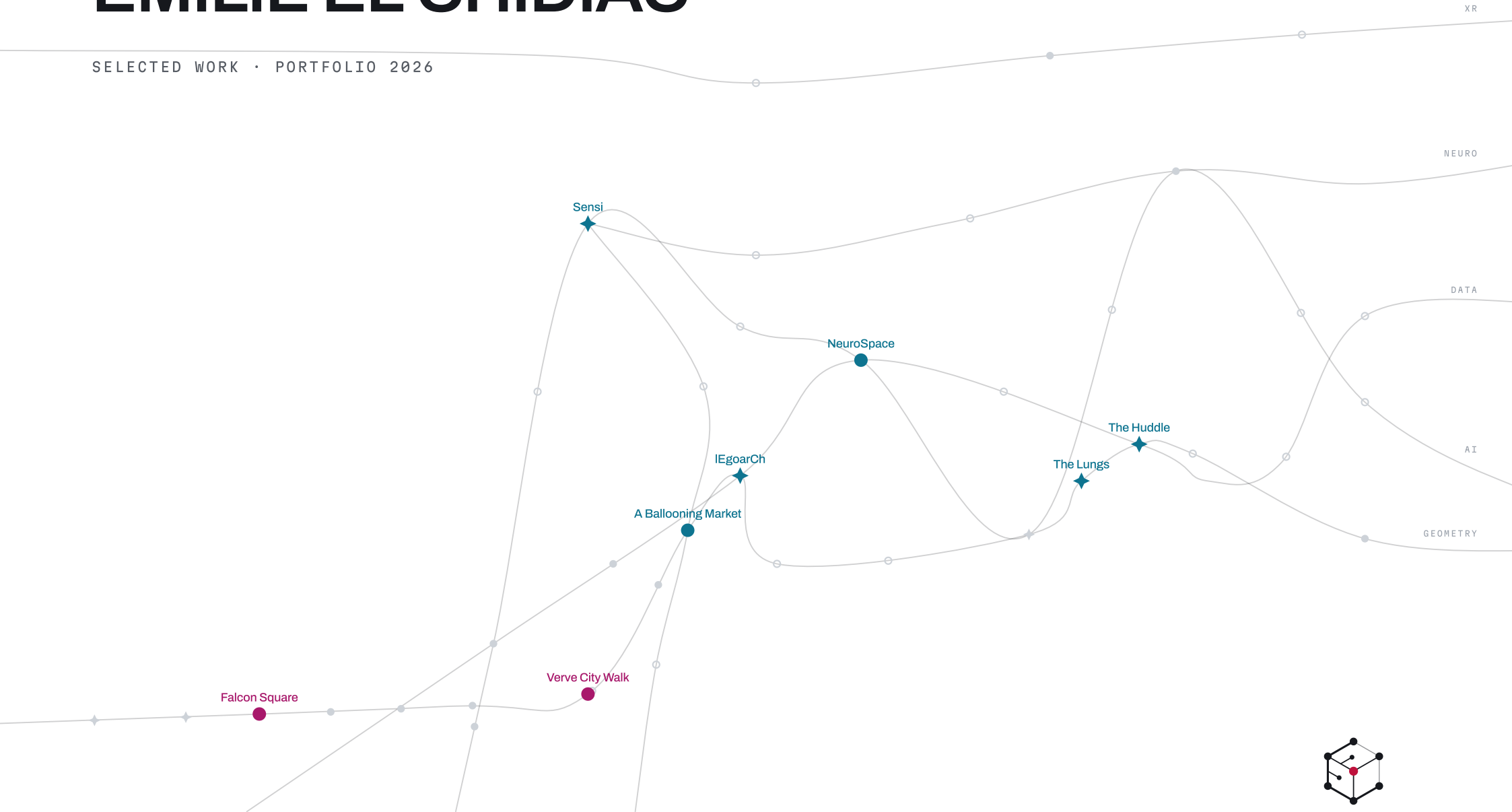


EMILIE EL CHIDIAC

SELECTED WORK • PORTFOLIO 2026



emiliechidiac.com

About

Emilie El Chidiac. I work with design, technology and minds.

WHO

I will not pretend I can fill a blank notebook from nothing. What I am good at is judging what the machine hands back and steering it. Someday that might be for a cursor instead of a person. Dystopian or utopian?

We build for the eye sometimes, which helps. Mostly we build for function, which is counterintuitive isn't it? We should be building for us to function at our best, not the building.

So I build the instruments: Sensi scores a floor plan across six senses, NeuroSpace estimates how a room reads to the brain and shows its weights so you can disagree with them.

It was never about being the most interesting person in the room. But the most interested one: what is x doing to y ?

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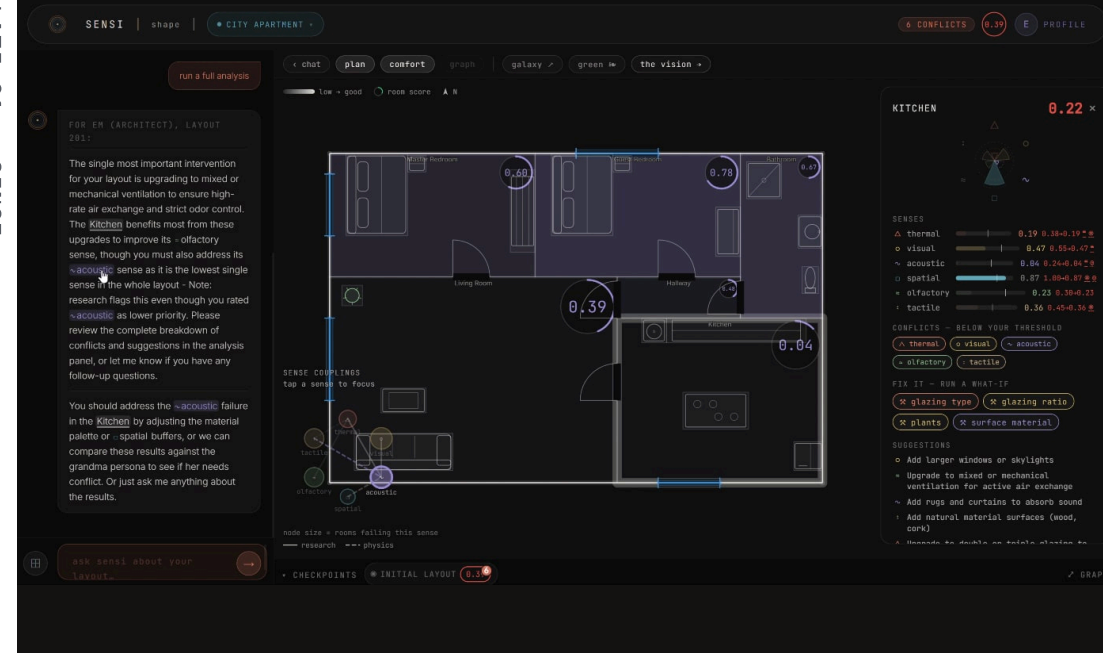
■ COMPUTATION & RESEARCH

MACAD STUDIO • TEAM OF 4

+ MACAD AWARDS 2026 • DESIGN COPILOTS • WINNER

Sensi

Comfort, designed on purpose: a copilot scores a plan across six senses, calibrated to a person, not an average.



WHAT

Every tool in the stack could tell us how a building performs. None of them would say how a room feels. Sensi closes that gap: a copilot that reads a floor plan and scores comfort across six senses (thermal, visual, acoustic, spatial, olfactory, tactile), calibrated to one person at a time, not an average. Project lead, A to Z, built with a team of four: Lakzhmy Mari Zaro, María Sánchez Domínguez, Charles Abi Chahine, and me.

WHY

Comfort is usually the thing we hope shows up after the design is done. I wanted it to be a layer you can interrogate while the plan is still soft, because you do not walk into a room and average your experience: the thing that is wrong is the thing you notice.

HOW

Onboarding calibrates the copilot to the person in the room, their thermal grudges, their noise tolerance. Comfort without a subject is just weather.

One action classifier, a single LLM call per turn, routes each request through a LangGraph state graph: analyze, edit, preview, audit.

A coupling matrix ripples every change into the neighbouring senses, so a fix that quietly breaks another score gets flagged, not hidden. *n.b. the six scores argue like a family. the coupling matrix is the dinner table.*

The copilot suggests edits the layout can absorb, previews them without committing, then renders the comparison and hands over the report.

WHAT CAME OF IT

We benched the judgment before trusting it: two LLM providers scored the same three scenes end to end, including a living room arranged to fail. They mostly agreed, and it would have been easy to call that validation. We wrote agreement is not truth into the notes instead, and kept every disagreement as data.

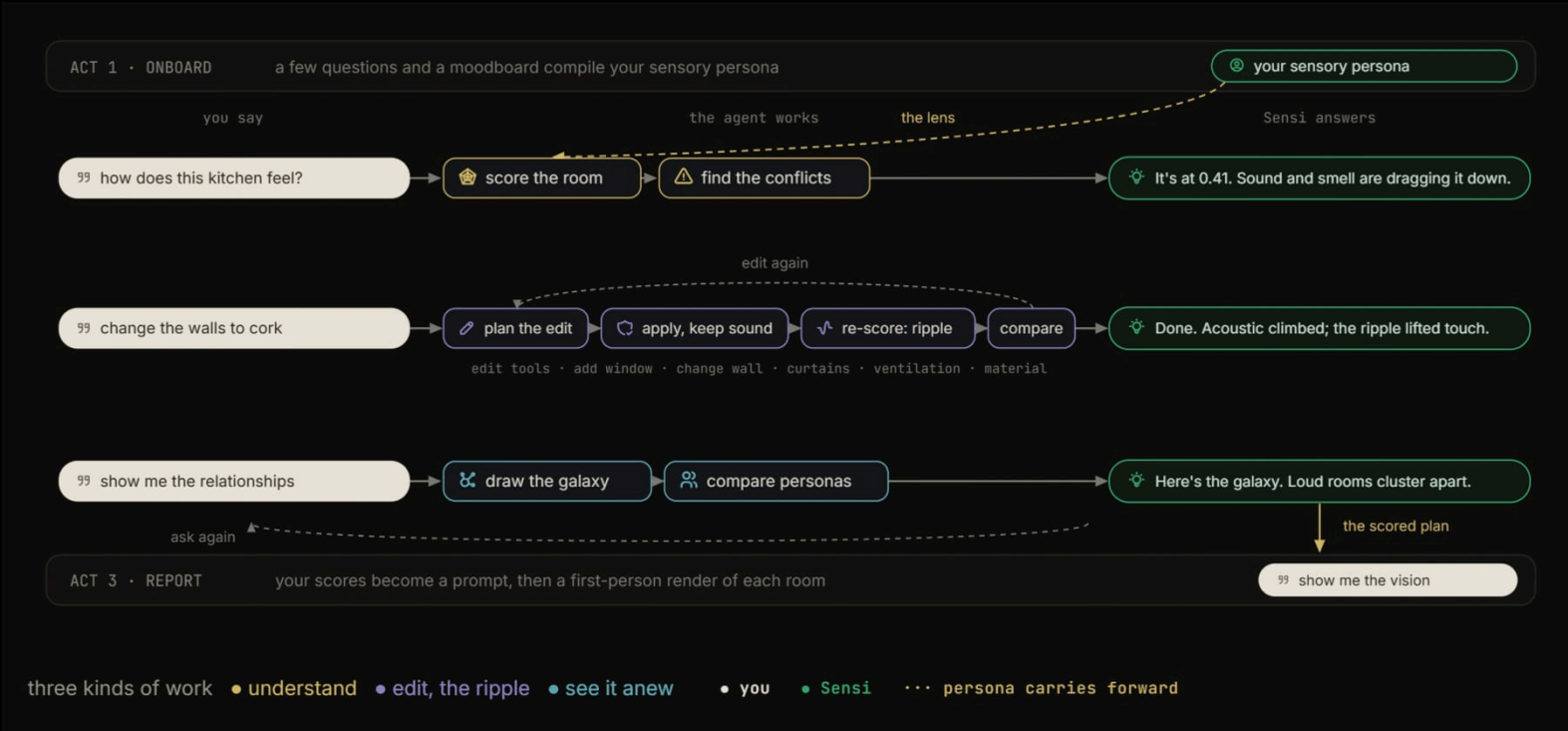
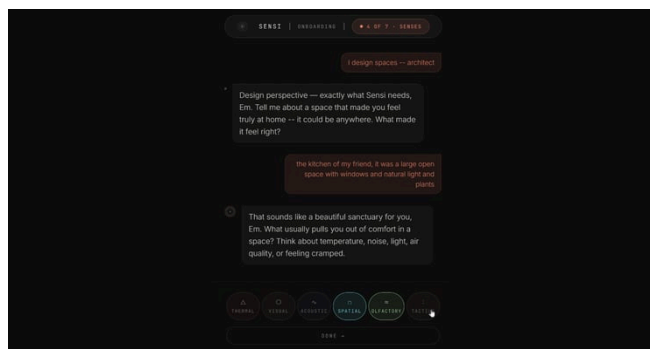


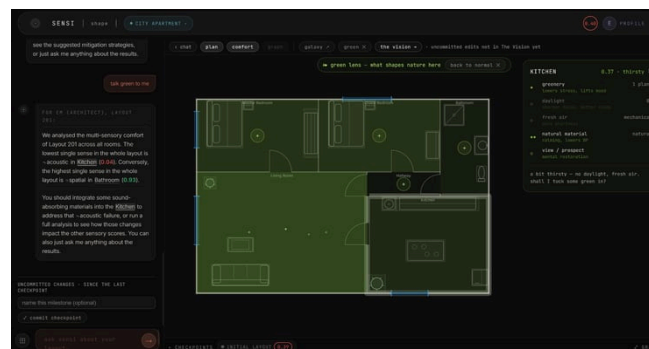
PLATE 01
MACAD STUDIO · TEAM OF 4

PYTHON · LANGGRAPH · FASTAPI
· REACT

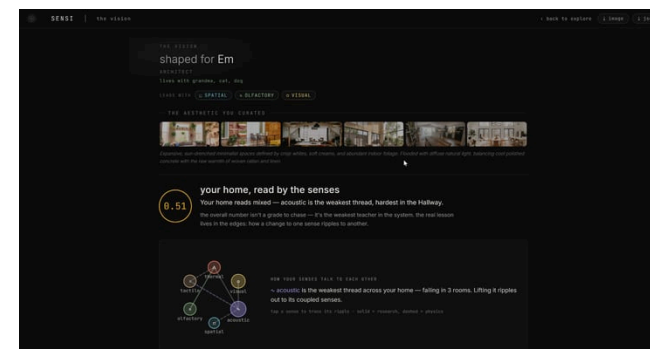
Act 2 of the pipeline as a flowchart: score, edit, ripple and galaxy loops around the plan while it is still soft



Sensi's onboarding chat calibrating the comfort copilot to one person: their noise tolerance, their thermal grudges



The green lens over a city apartment plan: what shapes nature here, the kitchen flagged thirsty for daylight and plants



The vision report Sensi hands over: comfort scores shaped for one person, with the edits the layout absorbed

■ COMPUTATION & RESEARCH

MACAD GENERATIVE AI · WITH CHARLES ABI CHAHINE

+ JURY AWARD

lEgoarCh

A render is only a promise until the bricks fit: AI imagines the set, code makes it actually buildable.

WHAT

A render is a promise, not a product: you cannot snap a JPEG together on your living-room floor. lEgoarCh takes a text prompt and returns a LEGO Architecture set that is digitally verified buildable: AI imagines it, deterministic code makes it snap together, brick by brick, out of real catalog parts. Built with Charles Abi Chahine, end to end as a pair.

WHY

The inspired gesture is cheap now; anyone can generate a thousand renders before lunch. The interesting problem moved to verification: what does it take to turn a generated image into something that provably holds together?

HOW

A LoRA tuned on a 40-image dataset teaches FLUX the LEGO Architecture look; the prompt becomes a styled render.

TRELLIS lifts the render into a 3D mesh, and the mesh is voxelized into brick space.

An optimizer places real catalog bricks into the voxels, enforcing connectivity, support, and perceptual color accuracy.

The set exports as LDraw, the format the brick world already speaks.

WHAT CAME OF IT

The most instructive moment was a failure: an intermediate model came back connected and supported, yet not legible as architecture. Structurally sound and visually wrong is still wrong, so legibility joined the constraints. The final pipeline was benched on three buildings, and the jury gave it their award.

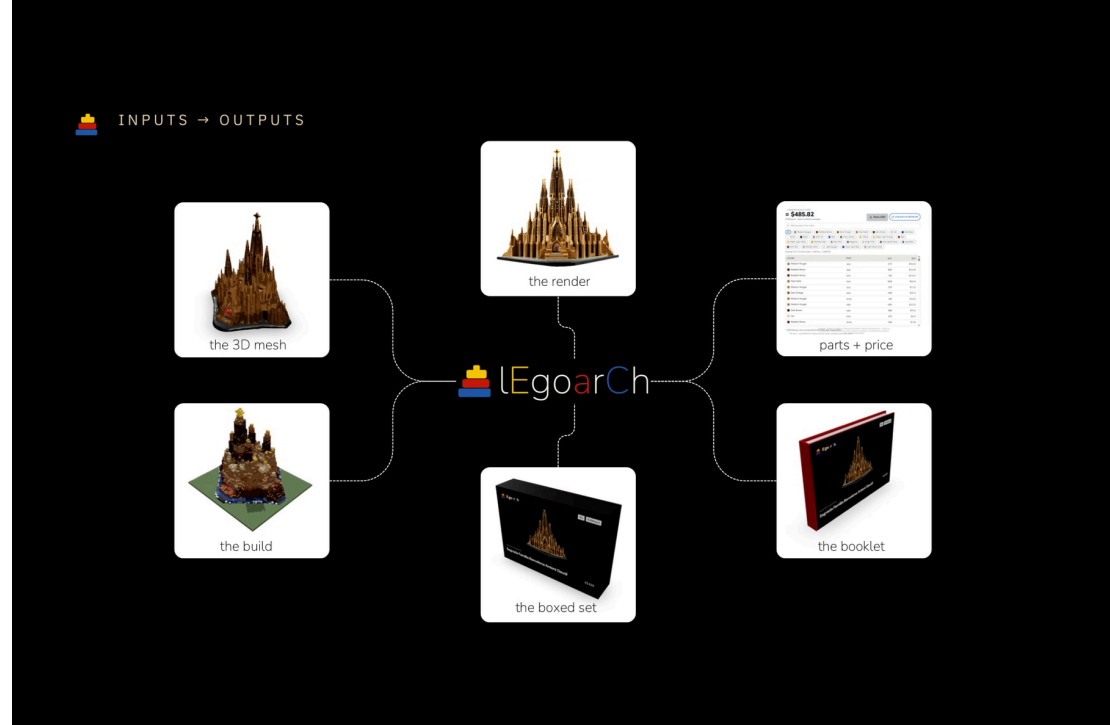
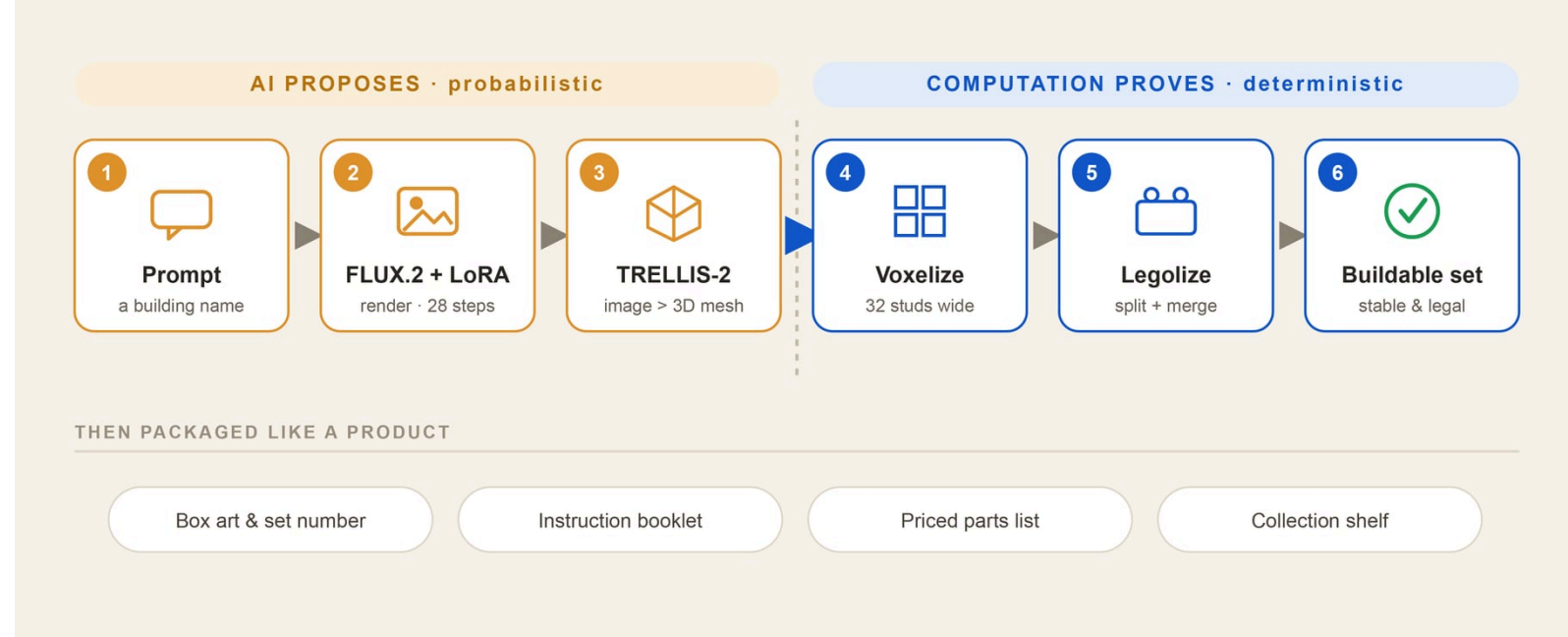


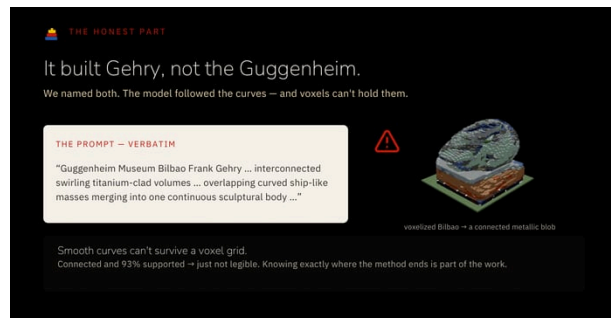
PLATE 02

MACAD GENERATIVE AI · WITH
CHARLES ABI CHAHINE

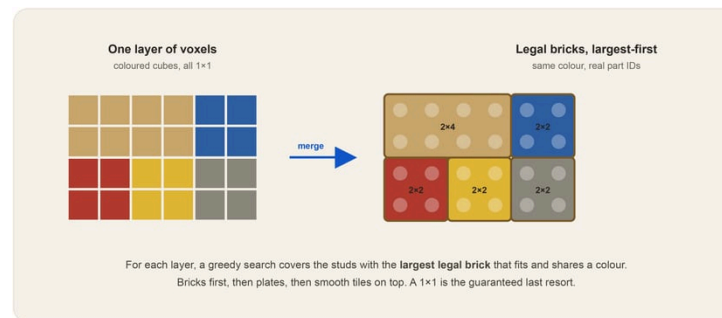
FLUX.2 KLEIN · TRELLIS-2 · LORA
· LDRAW



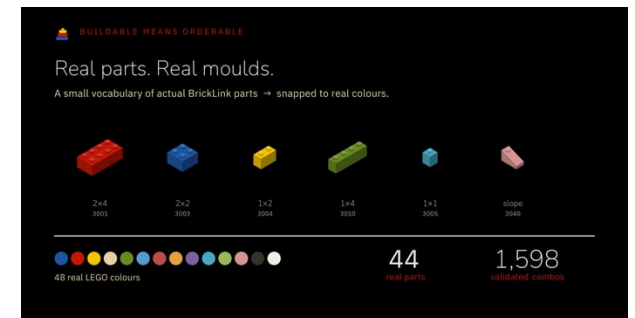
The lEgoarch GenAI pipeline: FLUX and TRELLIS propose, deterministic code verifies the LEGO set as digitally buildable



The instructive failure: the Guggenheim prompt returned a voxel blob, structurally sound but not legible as architecture



The split and merge pass packing the voxel grid into the largest legal bricks, fewer parts holding the same shape



Real parts and real moulds: 44 BrickLink parts in 48 colours, the catalog the optimizer is allowed to build from

NeuroSpace

Your room is doing something to you right now: move a slider and watch a browser score it live.

WHAT

You are sitting in a room right now, and its defaults are quietly working on you: the ceiling height nudging your cortisol, the daylight setting your circadian clock. NeuroSpace makes that invisible layer legible: move a slider and the room rebuilds while a score answers back, live. I built it on my own: a Grasshopper definition doing the heavy geometry on the server through Rhino.Compute, Three.js drawing the room in the browser, and a scoring pass that estimates the behavioral effect the moment you let go.

WHY

This is the thesis I keep circling: BIM, reframed from Building Information Modeling to Behavior Information Modeling. The information that matters is not just what a building is made of; it is what the building is doing to the person inside it.

HOW

Describe the room as parameters, not geometry: ceiling height, wall count and curvature, openings, organic form, plants. Every one is a slider.

Send the parameters to a Grasshopper definition on the server; Rhino.Compute evaluates it and streams the heavy geometry back, so the browser never has to model anything itself.

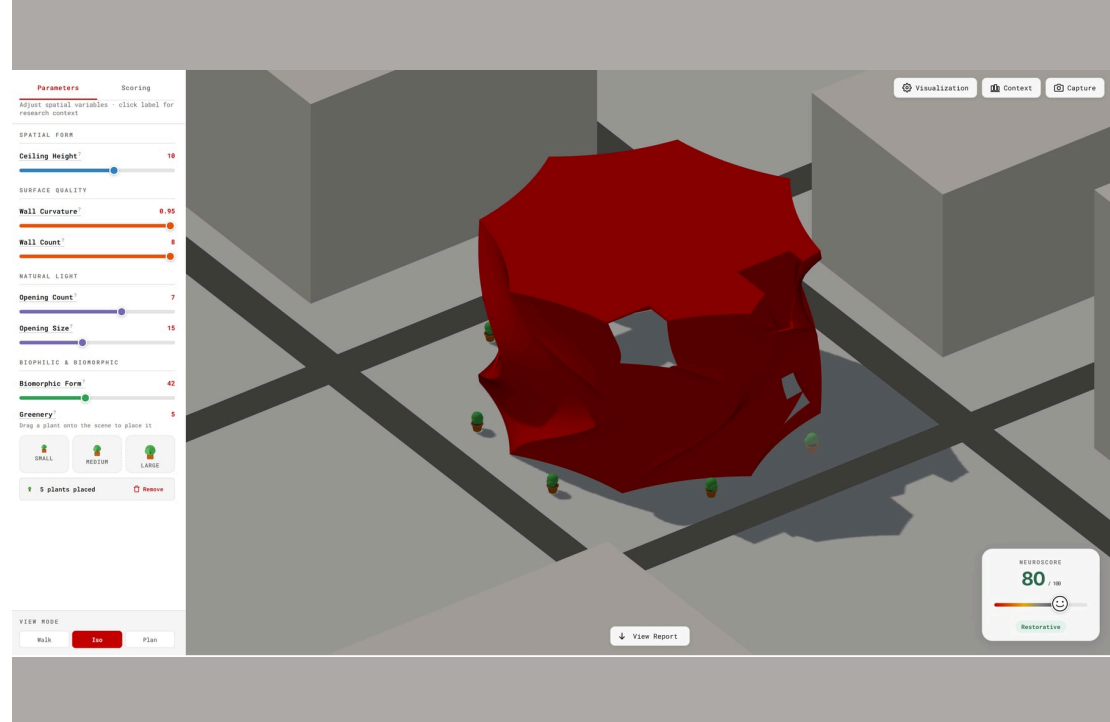
Draw the returned room with Three.js. Geometry is the slow path; it only recomputes when the shape actually changes.

Score the behavior on the fast path, in the browser, the instant a slider settles: a transparent weighted sum over the dimensions the research cares about. No server round trip, so the number answers as fast as you can drag.

WHAT CAME OF IT

The score is a heuristic, not an instrument. It leans on neuroarchitecture research and turns it into weights you can read in the public repo, which means you can also argue with them. *n.b. a score you can argue with beats a number you have to trust.* It estimates; it never measures your body, and it shows every assumption on the way to the number so you are free to overrule it.

PLATE 03 · NEUROSPACE



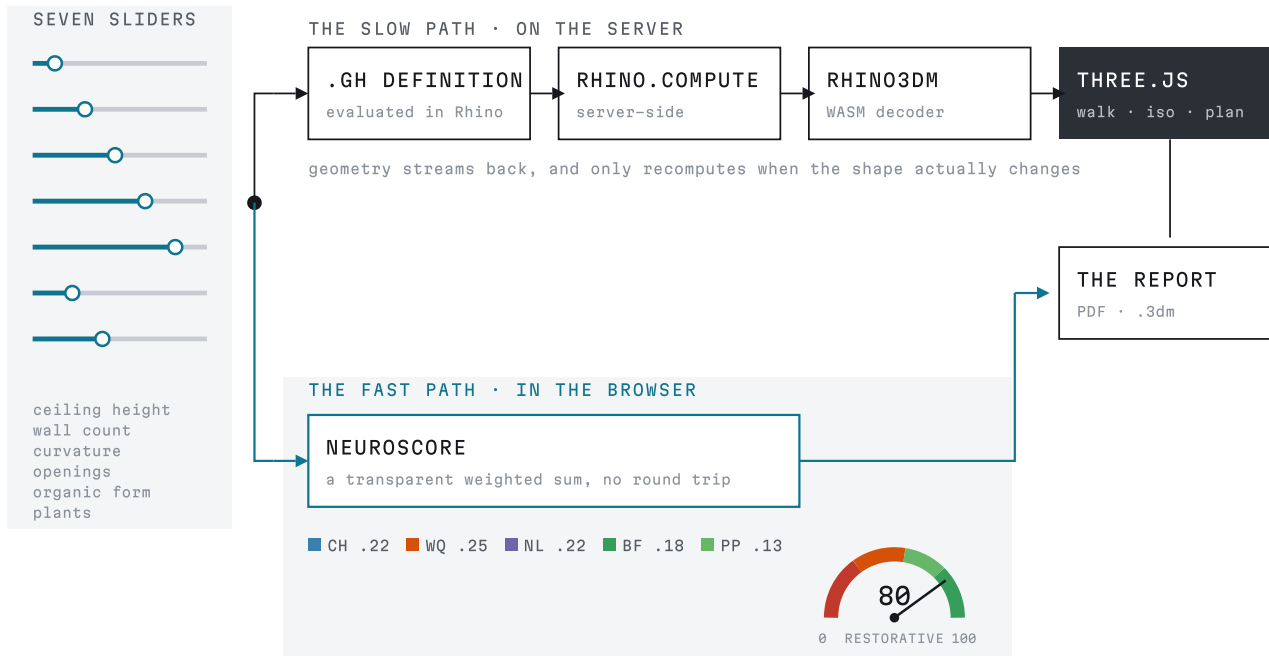
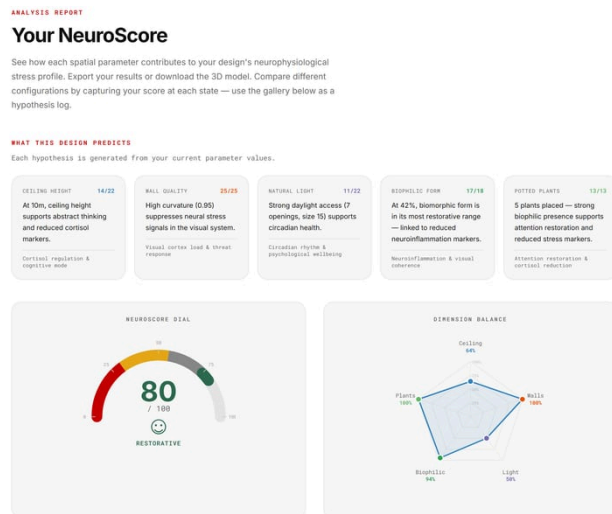


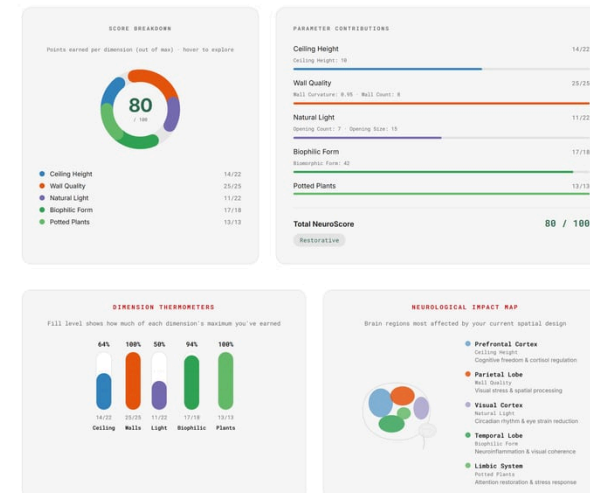
PLATE 03
MACAD · SOLO · LIVE APP

GRASSHOPPER · RHINO.COMPUTE · VUE 3 ·
THREE.JS

One slider move, two paths: the server returns geometry, the browser returns a score before you let go



Top of the NeuroSpace report: what this design predicts, hypothesis cards and the estimated wellbeing score dial



The report's breakdown: contribution bars and a brain impact map behind the room's estimated wellbeing score

■ COMPUTATION & RESEARCH

MACAD STUDIO • DATA TEAM OF 3 • LIVE APP

+ STUDIO AWARD

The Lungs

The studio's data was the architecture: a live app running a hyperbuilding designed to filter a city's air.

WHAT

Santiago has an air problem, the kind that sits in the valley like a guest who will not leave. The Lungs is a hyperbuilding designed to filter 12 million m³ of air a year, and the data team of three (María Sánchez Domínguez, Lakzhmy Mari Zaro, and me) built the live web app the whole studio ran on while designing it.

WHY

The studio's data was the architecture. Three design teams, ten weeks, one shared organism: somebody had to build its nervous system, and that turned out to be the most architectural thing in the room.

HOW

Build the platform in Vue 3, Vite, and Tailwind: KPIs, timelines, and team status, live for everyone instead of buried in spreadsheets.

Stream the 3D models in through Speckle, so performance data sits on geometry, not beside it.

Keep it running while the studio designs on top of it: the app was not documentation, it was the project's operating system.

WHAT CAME OF IT

The platform tracked three design teams across the full ten-week studio. Building a platform alongside the project it supports is messy, and the mess is the lesson: the building breathes, so does the app.

PLATE 04 • THE LUNGS

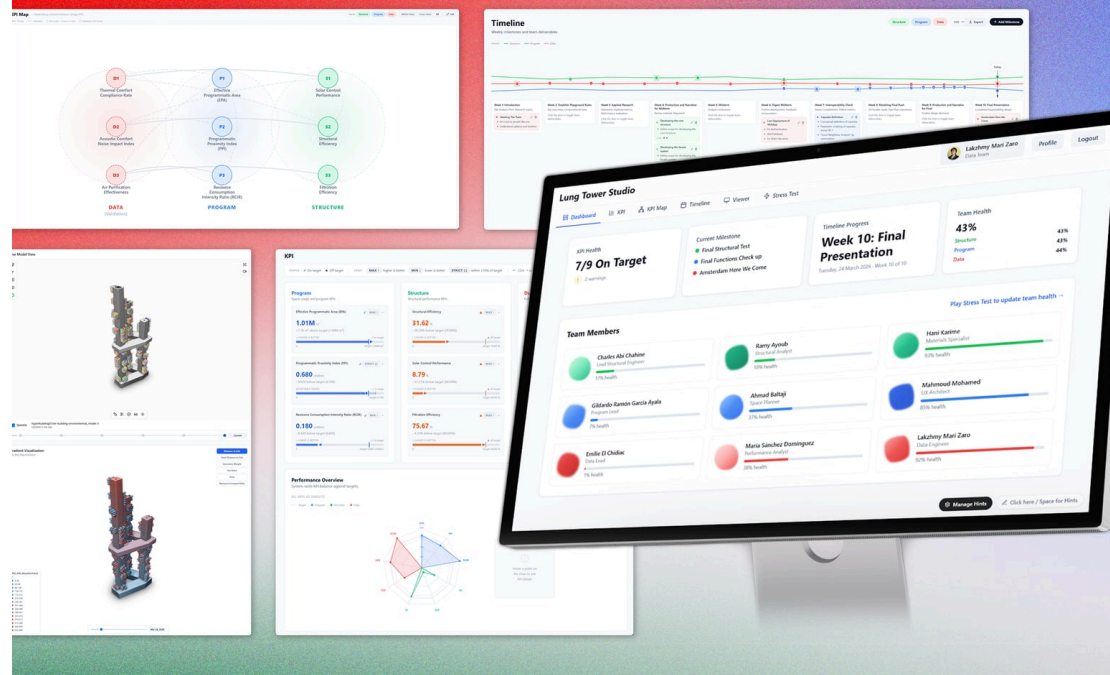
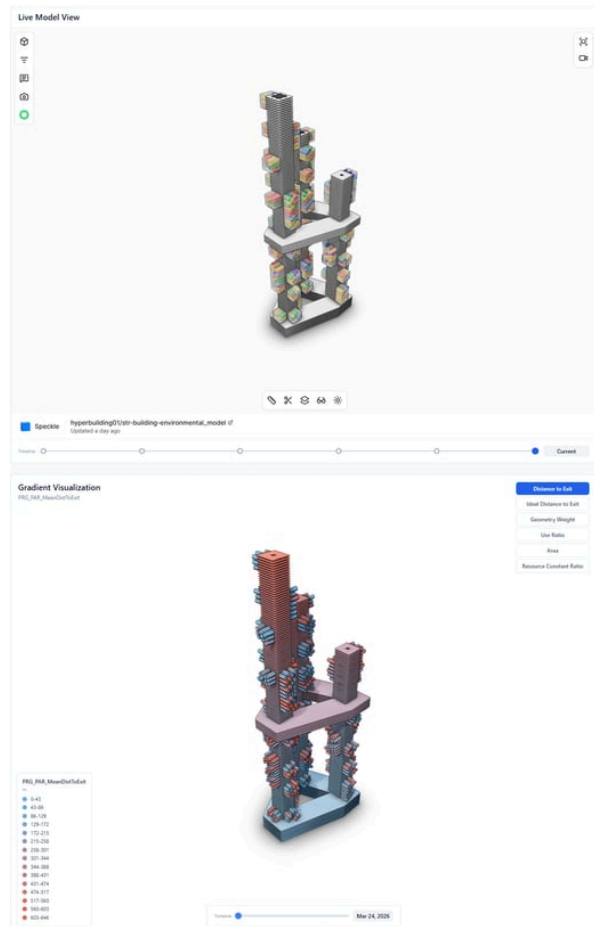


PLATE 04

MACAD STUDIO • DATA TEAM OF 3 • LIVE APP

VUE 3 • VITE • TAILWIND • SPECKLE



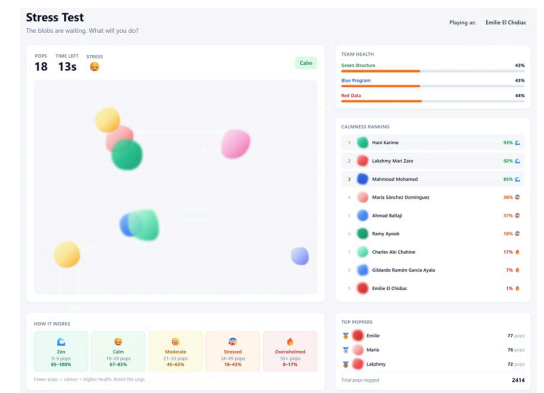
Speckle streams the design teams' tower model into the app, so performance gradients sit on the geometry itself



Ten weeks as a live timeline: three team tracks and the weekly deliverables the app kept honest



The KPI dependency network: data, program and structure indicators tied into one system the studio steered by



The stress test tab, a calmness leaderboard the studio actually played; fewer pops means calmer

■ COMPUTATION & RESEARCH

MACAD STUDIO · TEAM OF 4

+ STUDIO AWARD

The Huddle

Stop fighting the wind and build with it: modules that grow along the gusts instead of bracing against them.

WHAT

In Punta Arenas the wind never stops, so we stopped fighting it. The Huddle is a research and education hub grown from 4×4×4 m modules aggregated along the wind itself, wrapped in an envelope of three panel types (Shields, Lenses, Gills) that turns the facade into a readable map of climatic forces. A team of four, all hands on everything: María Sánchez Domínguez, Lakzhmy Mari Zaro, Charles Abi Chahine, and me.

WHY

Wind is the site's most abundant force, and the vernacular Kawesqar huts already knew what to do with it. Accepting the wind as a design partner, instead of a problem to brace against, turns the harshest constraint into the organizing logic of the whole building.

HOW

WASP grows the module cluster along the wind patterns, solar exposure, and program, the way the Kawesqar huts huddled.

Kangaroo settles the layout; Alpaca4D checks the structural behavior of the result.

A Global Index algorithm distributes the three envelope panels, so the facade reads the climate back to you.

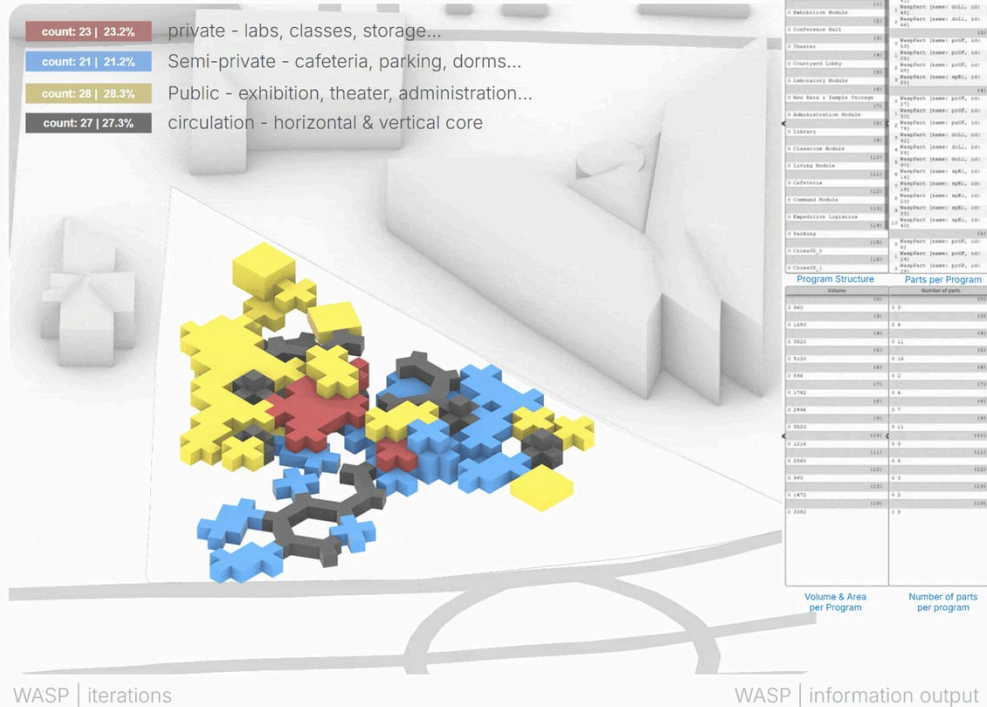
Vertical turbines sit in the aggregation's own wind tunnels, harvesting the force the form was grown from.

WHAT CAME OF IT

Because the building is an aggregation of known parts, the exact number of parts is known at any moment: the architecture stays buildable arithmetic, not a sculpture. Wind, once a problem, ends up the project's material.

PLATE 05 · THE HUDDLE





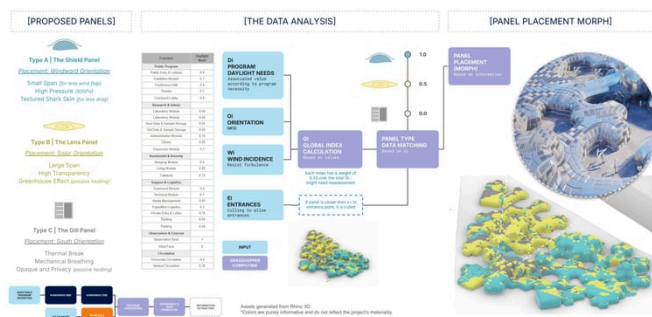
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PLATE 05

MACAD STUDIO • TEAM OF 4

WASP · KANGAROO · ALPACA4D

Animated WASP growth study: modules aggregating along wind and program across the Punta Arenas plot



■ COMPUTATION & RESEARCH MACAD · SOLO

A Ballooning Market

Physics is the difference between a mess and a roof: the balloons ghosted through each other until Kangaroo gave them awareness.

WHAT

A historic Cairo market receives a new roof of pressure-packed balloons that borrows the existing steel frame without modifying it. *n.b. the frame is the client, the balloons are the tenants, the solver is the lease.* Each balloon is a Kangaroo body with collision, inflation, and anchor goals; the settled cluster is meshed with Dendro and lit through CMY membranes in D5.

WHY

A building like Bab al-Luq usually gets offered two futures: a museum piece that can never be touched, or a cold glass-and-steel box dropped in the middle and called modern. Pneumatic parasitism is the third option: a soft new life that hugs the old bones instead of replacing them.

HOW

Trace the frame of Bab al-Luq; mark the nodes that can host anchors.

Seed balloons in the volume; declare radii, no physics yet.

Add collision and inflation goals; anchor the cluster; let the solver settle.

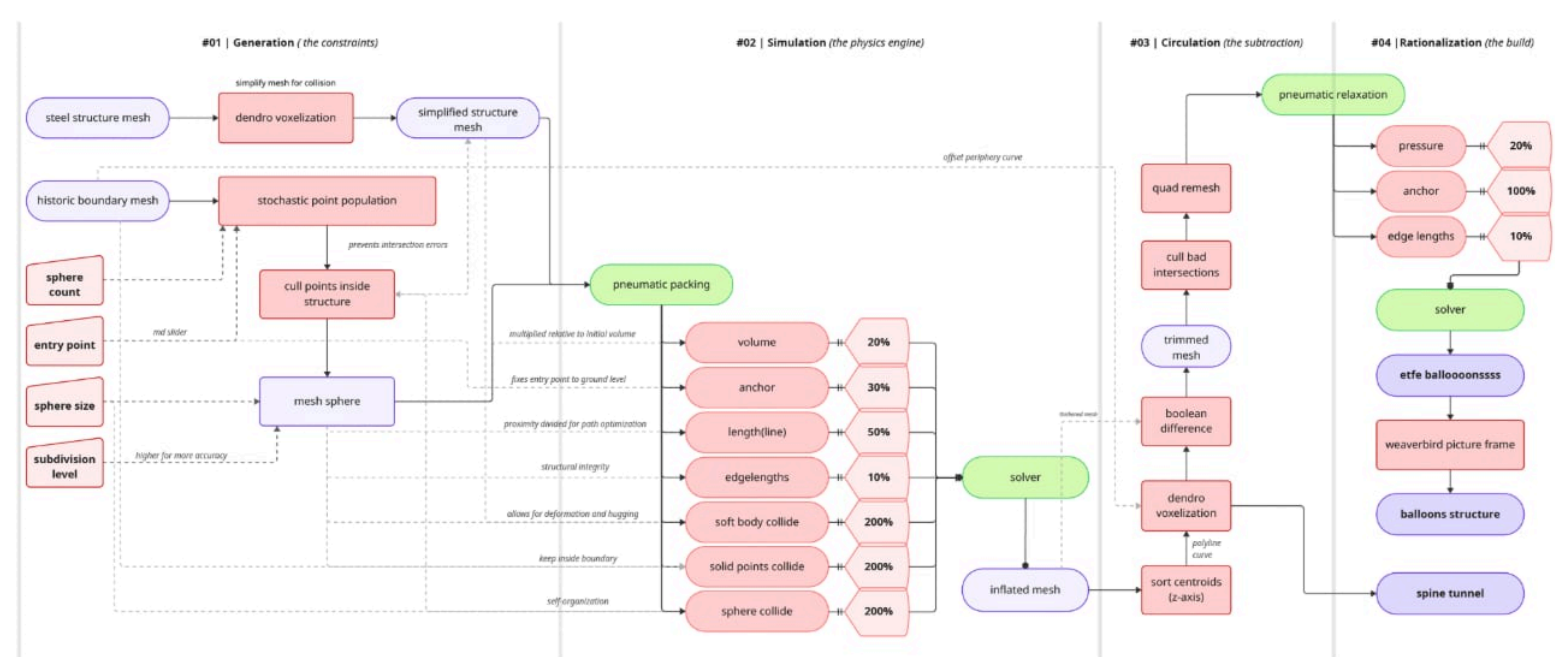
Mesh the settled cluster with Dendro; render daylight through the CMY membranes in D5.

WHAT CAME OF IT

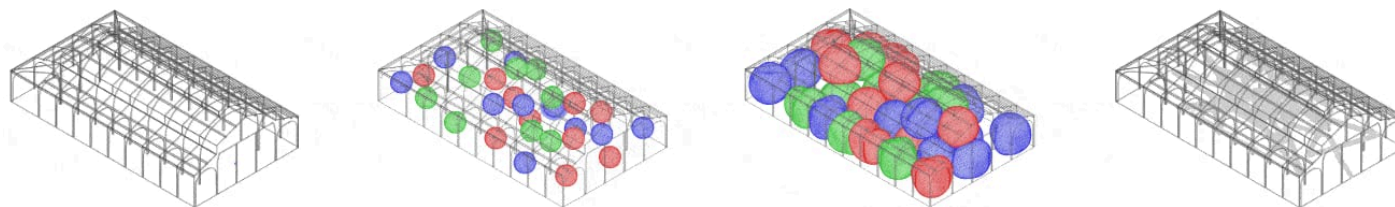
The first run failed in an instructive way: the balloons had no physical awareness of each other, just ghosting through one another in a chaotic, colorful mess. The fail stayed in the record on purpose; people read honesty faster than polish. And the tuning became the craft: collision too low and they cuddle, too high and they panic. You tune it like a thermostat.

PLATE 06 · A BALLOONING MARKET

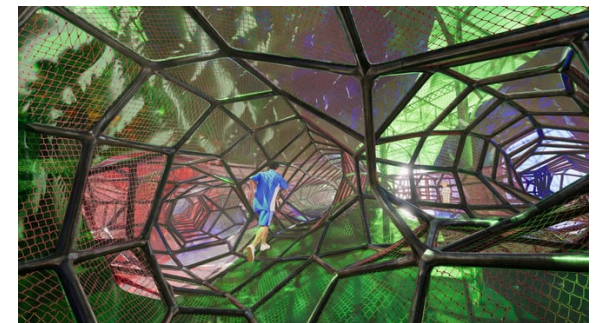




The Grasshopper algorithm behind the inflation simulation, from constraints to Kangaroo goals to the settled cluster



Four stages of the Kangaroo inflation: seeding the balloons, anchoring, settling, and threading the spine path



Render of the playscape tunnel through the balloon field, a kid running the voronoi net spine

◆ DESIGN & PRACTICE

SOMA · DESIGN ARCHITECT · 2023–24

Verve City Walk

Verve at City Walk, from parametric facade studies to the BIM set, with three more Dubai towers behind it.



WHAT

Verve City Walk is the two tower high-rise SOMA designed at City Walk in Dubai: a shared amenities podium under both towers, balconies cut into the facade rather than hung off it. As a design architect at SOMA I ran design exploration and parametric facade studies in Rhino and Grasshopper, and carried them into the Revit BIM model for floorplans, facade strategies and interior layouts of the residential towers. The same year carried three more Dubai towers, District O, Enara and Saria, from massing studies into their BIM sets.

WHY

These are the years where I learned that a parametric study only matters if it survives contact with a drawing set. Practice is where the elegant definition meets the door schedule, and both have to win.

HOW

Concept stays in Rhino and Grasshopper, where the massing can still change its mind: plot, podium, unit mix, all of it still cheap to move.

Design development, and everything moves into Revit: landscape planning, the typical floor plates, the interior layouts of a luxury tower. This is where gross floor area starts running the project.

Revit families built with the visualization team, and the same model going out three ways: 3D prints, renders, a VR walkthrough in Unreal.

Then every consultant back into the file, structure, MEP, facade, because the Revit model is the submission and it has to be true on the day it leaves.

WHAT CAME OF IT

At university nobody pays for the square metres, so nobody counts them. Then you are in practice and gross floor area is the only number in the room: every balcony cut into the facade, every core shifted, every planted terrace, all of it weighed against that one figure. I used to think that was the boring part. It turns out to be the part that decides what gets built. What came of these years is the coordination more than any drawing: everyone working into one model, and a client putting on a headset to walk a tower that did not exist yet.

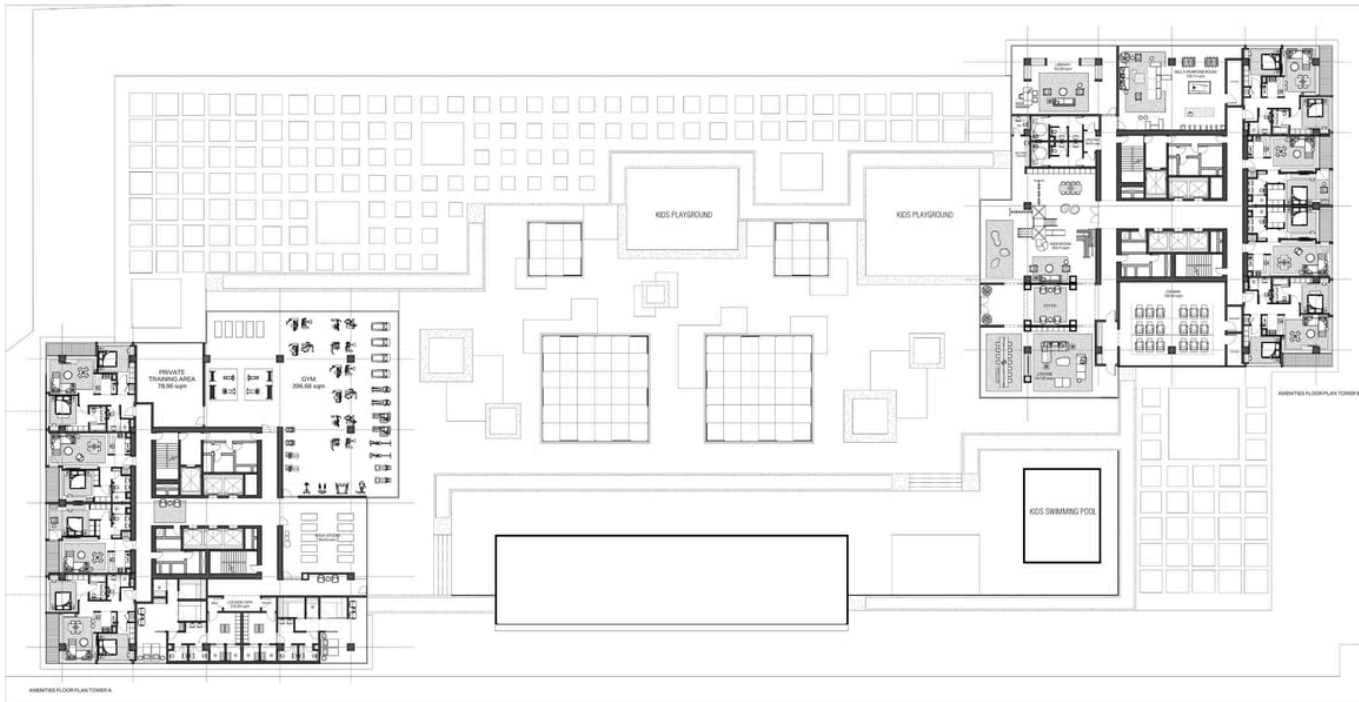
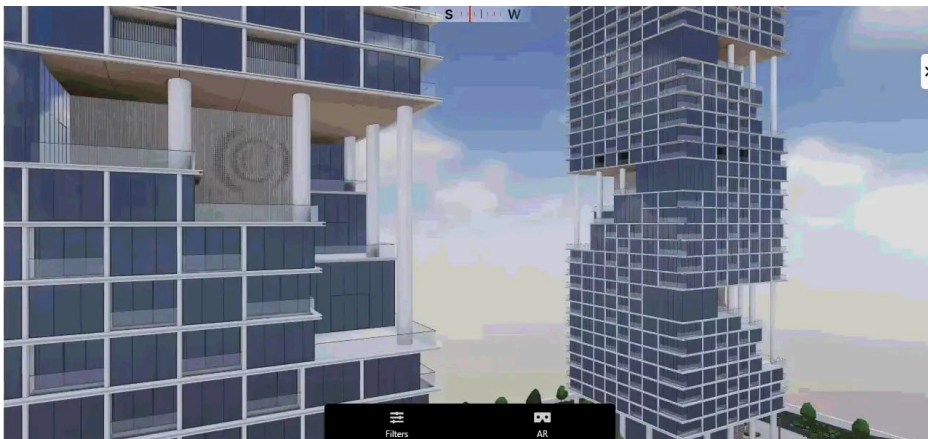


PLATE 07

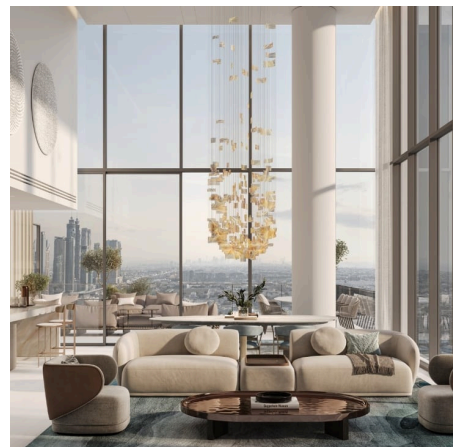
SOMA • DESIGN ARCHITECT • 2023-24

RHINO • GRASSHOPPER • REVIT

Amenities level plan for the two Verve towers: gym, spa, yoga studio, cinema, kids pool and playgrounds on the podium

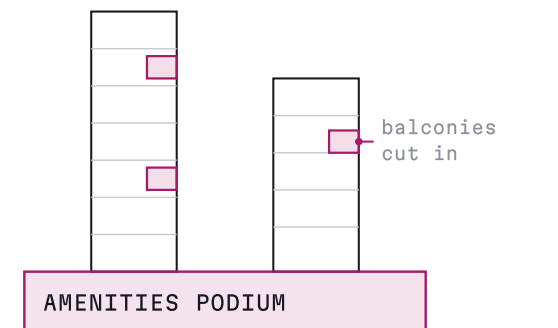


Screen capture of the interactive 3D unit selector orbiting the twin Verve towers, one unit footprint highlighted



Interior render of a double height Verve lounge, a suspended chandelier and floor to ceiling glass over the skyline

TWO TOWERS, ONE PODIUM



gym • spa • yoga • cinema
kids pool • playgrounds

Two towers of different height over one shared amenities podium, with the balconies cut into the facade rather than hung off it

◆ DESIGN & PRACTICE

JEMMA CHIDIAC ARCHITECTS • 2022

Falcon Square

An aircraft's takeoff lines, mirrored twice into a steel falcon: a monument for Al Khobar, led from the first sketch.



WHAT

A monument for a highway roundabout in Al Khobar: MIDAN AL SAKR, the office's Moujassam Watan competition entry. Layered steel blades form a falcon about to take off, its two wings carrying pierced Arabic calligraphy over a roundabout re-landscaped in stone wedges, lawns and water. As an architectural designer at Jemma Chidiac Architects, I started the concept sketches and led the design from brainstorming to the proposal, with the model built in Rhino SubD and the visualizations and animations rendered in-house.

WHY

The concept began with my ink studies of an aircraft: takeoff angles, control surfaces, the lines a plane draws when it leaves the ground. Mirrored once, the drawings opened a void; mirrored again, they closed into a falcon, the national bird carrying the national script. A monument for a roundabout has one job, to be read at speed, and a takeoff line is the fastest line there is.

HOW

Sketch the aircraft in ink: takeoff pitches, primary and secondary control surfaces.

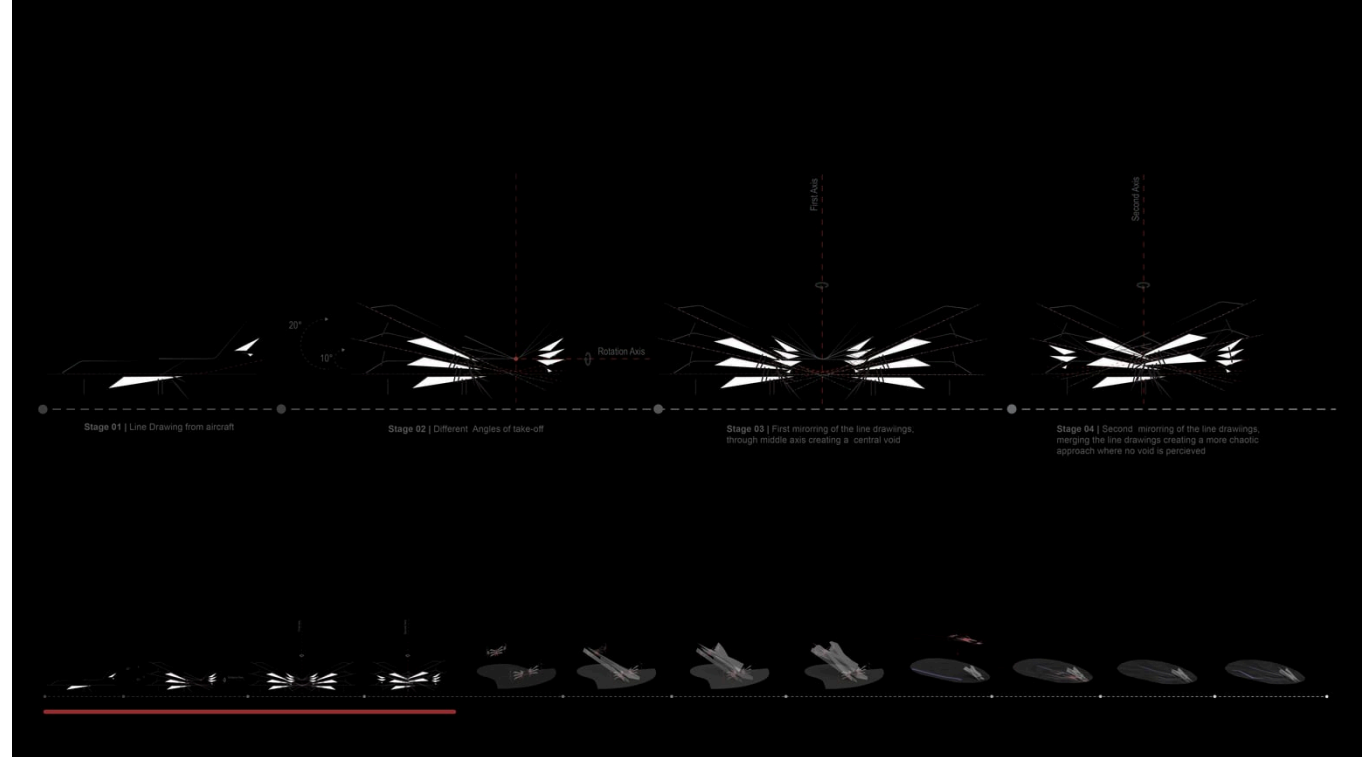
Layer the takeoff traces at different angles, then mirror the set twice, first opening a central void, then merging the drawings until no void is read.

Build the layered blades in Rhino SubD, thread the pierced calligraphy panels between them, and re-landscape the roundabout so the ground carries the bird.

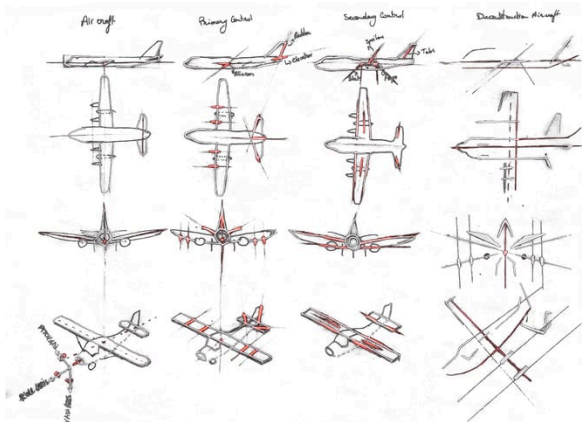
WHAT CAME OF IT

The proposal shipped as the office's competition entry and lives on the practice's site. The falcon stayed on the drawing board, which is where most competition monuments live; the sketches that started it are in the gallery.

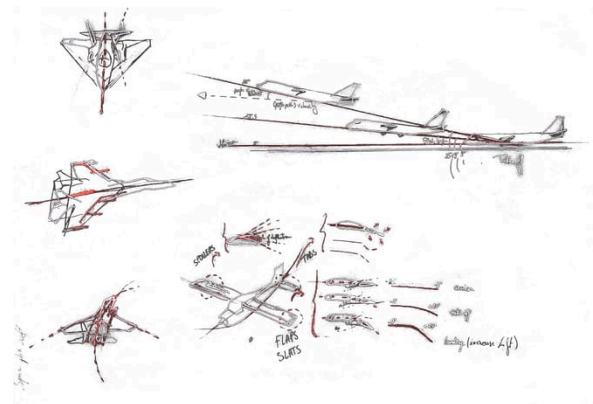
RHINO • SUBD • ARCHVIZ



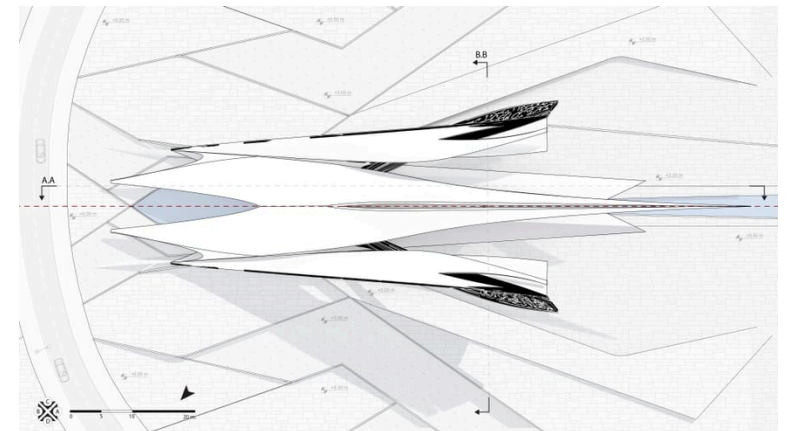
An aircraft take-off line, mirrored twice, becoming the falcon



Sketches studying aircraft anatomy: control surfaces, flaps and spoilers, and the pitch, roll and yaw axes



Sketches of jets and a take-off diagram, tracing climb angles, flaps and slats that seed the monument's lines



Site plan of the falcon monument on the roundabout, mirrored wings on one axis with water channels and section lines

Index



■ Sensi
MACAD · +



■ IEgoarCh
MACAD · +



■ NeuroSpace
MACAD



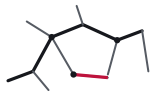
■ The Lungs
MACAD · +



■ The Huddle
MACAD · +



■ A Ballooning Market
MACAD



■ Encoding Urban Risk
MACAD



▲ Cappelletti Pavilion
MACAD



■ Optimizing for the Mind
MACAD



■ Narkomfin as a Graph
MACAD



■ Data into Geometry
MACAD



■ Tsukiji Fish Market
MACAD



▲ Chair Simulation
MACAD



▲ Astroidal Ellipsoid
MACAD



▲ A Playscape
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▲ The Homage
LAU · +



◆ Verve City Walk
SOMA



◆ Rings of Mars: Ring
4000
SELF · +



◆ The Encounter
JEMMA · +



◆ Falcon Square
JEMMA



▲ XR for Education
LAU

THE THOUGHTS, LATEST FIRST

○ <i>explaining things</i>	JUL 2026	○ <i>adjacency is not access</i>	JUN 2026	○ <i>buildings that respond</i>	JUN 2026
○ <i>experiences are data</i>	MAY 2026	○ <i>comfort as data</i>	MAY 2026	○ <i>when the tool scores people</i>	APR 2026
○ <i>latent space</i>	APR 2026	○ <i>drawing as interface</i>	MAR 2026	○ <i>behavior information modeling</i>	JAN 2026



this is what's on my mind

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